

JORDAN NEWMAN

Software Architect | Distributed Systems, Infrastructure, Security, DevOps

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Selected public repositories are available. The most significant engineering repositories are private; access is available upon request.

SUMMARY

Software architect and hands-on systems engineer with experience designing production software and infrastructure across CDN, hosting, enterprise software, security, backend platforms, and DevOps. Architected systems operating across approximately 3,000-4,000 servers, 10 CDN locations, and more than 65 Gbps of peak traffic. Combines technical leadership with deep implementation experience in C, FreeBSD, networking, kernel development, reliability, and performance engineering.

TECHNICAL SKILLS

- **Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic
- **Systems:** FreeBSD, Linux, kernel modules, system calls, operating systems, kqueue, sendfile(), GDB, TCP/IP, non-blocking I/O
- **Networking/Security:** HTTP, DNS, CDN architecture, socket programming, DDoS detection/mitigation, packet capture, libpcap, IPFW, authentication, telemetry
- **Cloud/DevOps:** Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Architected the custom HTTP serving and CDN software platform for infrastructure spanning thousands of servers and multiple points of presence.
- Designed high-availability, DDoS mitigation, traffic analysis, telemetry, authentication, and operational-control systems.
- Created a kernel-aware event-driven server architecture using kqueue, sendfile(), non-blocking sockets, I/O workers, and FreeBSD-specific optimizations.
- Designed a transparent FreeBSD kernel pathname-virtualization layer for multi-tenant hosting without application changes or chroot jails.
- Standardized Kubernetes, Terraform, Azure DevOps, CI/CD, and security-scanning practices across engineering teams.

SELECTED SYSTEMS PROJECTS

- **PrimeHTTPD:** Architected and implemented a high-performance, non-blocking, kqueue-based HTTP/CDN server in C for FreeBSD. Used sendfile(), SF_NODISKIO, TCP_NODELAY, TCP_NOPUSH, O_NONBLOCK, accept_filter_http, persistent connections, and dedicated I/O workers.
- **VirtualDir:** Designed and developed a FreeBSD kernel module that intercepted filesystem-related syscalls and transparently remapped paths from configuration, enabling shared physical layouts without chroot jails; included the vdcli runtime management tool.
- **PrimeDump:** Built a libpcap-based real-time traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 with ncurses views and IPFW integration.
- **PrimeDNSTop:** Built a DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source and domain activity.
- **TAFOS:** Developed an educational x86 operating-system kernel in C and assembly with a custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist

Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Served as a member of the Architecture Team, collaborating with fellow architects and the architecture lead to make technical decisions, define engineering standards, guide platform direction, and develop shared libraries, documentation, and AI-assisted engineering workflows.

DDI Systems - Senior Development Specialist

Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET components in C# and integrated Microsoft Office 365 SMTP OAuth for secure customer-facing workflows.

ISPrime LLC - CEO

Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - April 2018

- Built ERP integrations and automated order, catalog, image, SDS, SFTP, SellerCloud, Essendant, and MultiValue/Universe data workflows.

Too Much Media LLC - Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems and implemented OAuth 2.0 authorization for protected sites.

MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features and middleware for cross-platform database migration.

AJPM, LLC - Programmer / Owner

New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing, inventory, repair, resale, and financial workflow software.

ISPrime Inc. - CIO / Partner

Weehawken, NJ | March 2001 - December 2014

Infrastructure Leadership

- Architected and operated hosting and CDN infrastructure spanning approximately 3,000-4,000 servers, 10 locations, multiple datacenters, and more than 65 Gbps of peak traffic.

Systems Software Engineering

- Designed and built PrimeHTTPD, a high-performance, non-blocking, kqueue-based HTTP/CDN server in C for FreeBSD; deployed it across approximately 200 servers and supported more than 150,000 concurrent connections.
- Leveraged `sendfile()`, `SF_NODISKIO`, `TCP_NODELAY`, `TCP_NOPUSH`, `O_NONBLOCK`, `accept_filter_http`, persistent connections, and dedicated I/O workers to keep disk operations from blocking the networking event loop.
- Developed production FreeBSD kernel modules including VirtualDir, KeepClean, and StatCache.

Security and Networking

- Built DDoS detection and mitigation systems, packet and DNS analysis tools, monitoring, telemetry, authentication, deployment, and security software used across production infrastructure.

FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime and created automated availability and health alerting.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.